

MOHD KHALID

Software Engineer | Backend & AI

Distributed Systems • AI Systems • Intelligent Automation

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PROFESSIONAL SUMMARY

- Software Engineer specializing in backend systems, AI applications and intelligent automation, building production platforms serving **10,000+ users** and supporting **100,000+ travel bookings**.
- Designed distributed systems, AI-powered document intelligence and event-driven workflows that automated complex travel operations through scalable backend architectures.
- Founding Backend & AI Engineer with end-to-end ownership across architecture discussions, API design, implementation, deployment and production support in a fast-paced startup environment.

PROFESSIONAL EXPERIENCE

Flyo.ai

May 2024 – Present

Founding Backend & AI Engineer

- Collaborated directly with founders and CTO to design backend architecture, evaluate technical trade-offs and deliver production features across consumer and enterprise travel platforms.
- Engineered scalable backend services and APIs powering AI-enabled travel automation, itinerary management and enterprise travel workflows for **10,000+ users**.
- Designed an event-driven Gmail ingestion pipeline that synchronized historical travel emails before transitioning to incremental synchronization using Gmail Watch and History IDs for reliable, idempotent processing.
- Designed an event-driven document processing pipeline where Gmail Watch events triggered AWS Lambda for AI-powered document extraction before publishing normalized travel data to Amazon SQS for asynchronous persistence and downstream processing.
- Developed AI-driven document intelligence pipelines that transformed booking confirmations, passport images and travel documents into structured travel data using LLMs, OCR and prompt engineering.
- Fine-tuned GPT models for structured travel document extraction and developed automated evaluation pipelines to validate JSON outputs, measure extraction accuracy and support continuous model improvement.
- Developed supervisor-based multi-agent workflows using LangGraph to orchestrate specialized agents for booking, travel history, flight tracking and web check-in through modular tool routing.
- Designed a secure data-access layer for AI agents using PostgreSQL views, exposing only business-relevant fields through read-only abstractions instead of granting direct access to production tables.
- Engineered autonomous airline web check-in workflows combining browser automation, distributed scheduling, passenger document intelligence and asynchronous boarding-pass retrieval.
- Designed a dual-storage architecture using PostgreSQL for transactional data and MongoDB for storing unstructured AI processing artifacts, enabling traceability, debugging and prompt refinement.

KEY ENGINEERING SYSTEMS

Intelligent Travel Automation Platform

AI-powered itinerary intelligence

- Designed a unified travel ingestion pipeline consolidating Gmail, Google Calendar, booking confirmations, passport images and manual flight searches into structured user trips.
- Developed AI-driven document intelligence workflows that transformed unstructured travel documents into normalized travel data using LLMs, OCR and prompt engineering.
- Built proactive flight intelligence services integrating aviation APIs with WhatsApp and push notifications for real-time operational flight updates.

Autonomous Flight Check-in Engine

Browser automation

- Designed automated scheduling workflows that initiated airline web check-in based on airline-specific check-in windows without user intervention.
- Engineered browser automation workflows (Browser Use) capable of interacting directly with airline websites to complete passenger web check-in.
- Developed asynchronous boarding-pass retrieval pipelines that continuously monitored user emails after successful check-in and automatically delivered boarding passes through WhatsApp.

Enterprise Travel Operations Automation Platform

Enterprise AI workflows

- Built AI-powered workflows that classified enterprise travel emails into quotations, bookings, support requests and marketing communications while generating concise operational summaries.
- Designed intelligent orchestration pipelines that routed hotel and flight requests into specialized extraction workflows for downstream processing.
- Collaborated on building a subscription-based travel platform enabling agencies to automate customer operations, manage bookings, create SEO-friendly websites and streamline subscription management.

TECHNICAL SKILLS

Languages	JavaScript, TypeScript
Backend	Node.js, Express.js, API Design, REST APIs, Webhooks, OAuth 2.0, JWT Authentication
AI Systems	AI Agents, LLM Applications, LangGraph, Fine-tuning, LLM Evaluation, Prompt Engineering, Document Intelligence, Structured Data Extraction, OCR, LlamaParse
Distributed	Event-driven Architecture, AWS Lambda, Amazon SQS, Distributed Scheduling, Browser Automation, Browser Use, Webhook Integrations
Databases	PostgreSQL, MongoDB, Redis
Cloud	AWS EC2, AWS S3, New Relic
Integrations	Variflight, Gmail API, Google Calendar API, Outlook OAuth, WhatsApp Business API, OneSignal, DragonPass, Razorpay, Dodo Payments

EDUCATION

Masai School	2023 – 2024
Full Stack Web Development Bootcamp	
Jamia Millia Islamia	2018 – 2022
Bachelor of Technology, Electronics & Communication Engineering	